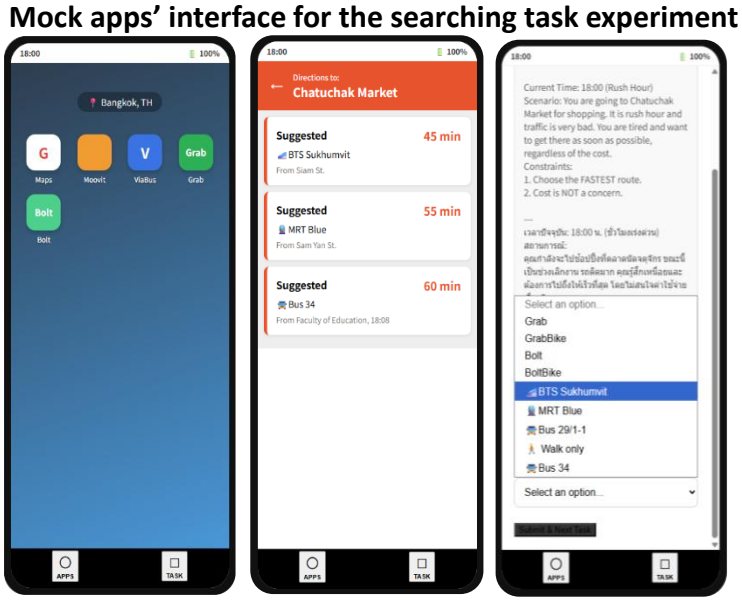
Mobility App Fragmentation and Information Searching Behavior in Developing Asian Cities: Evidence from Bangkok

アジア新興都市における交通アプリの断片化と
情報探索行動：バンコクの事例分析

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Mock apps' interface for the searching task experiment



Map of cities selected for worldwide app analysis



1 Background

Mobility apps and mobility information

- Widespread use of mobility apps improves urban mobility.

App fragmentation in developing Asian cities

- Uncoordinated digitalization causes app fragmentation
- App fragmentation = Individuals are frustrated by using multiple apps.
- **App switch** is the user's coping behavior to integrate fragmented information.

Research question

What type of users are affected by the fragmentation of mobility apps?

2 Goals

This study aims to:

(1) Compare the mobility app market worldwide to identify mobility app fragmentation in developing cities.

(2) Analyze heterogeneous route-searching behavior under fragmented apps.

(3) Discuss implications for a user-friendly app environment.

3 Methods

1. App Market Analysis

- App data collection from Google Play
- Mapping the app fragmentation
- Major cities in the world are compared.

2. Behavior Analysis

- Route Searching behavior task experiment on students in Bangkok.
- Model estimation of app switch behavior
- Unrelated Regression (SUR) model

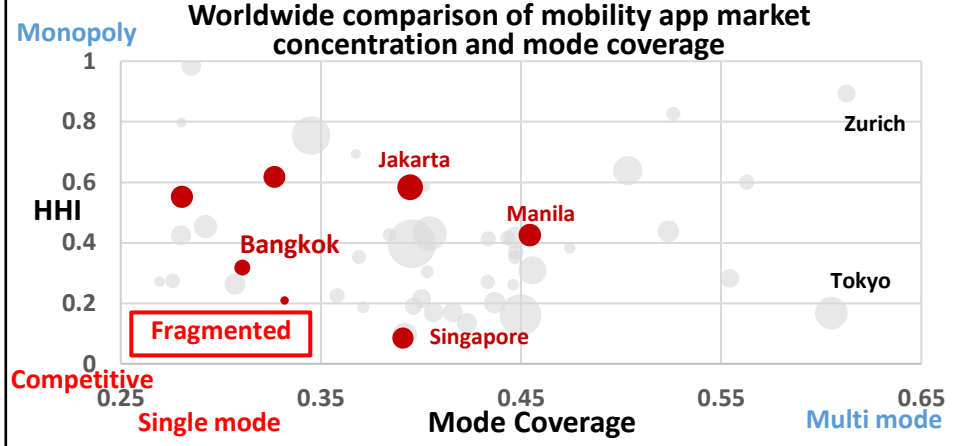
Discussion on policy implications

4 Results

1. App fragmentation situations in developing Asian cities

Southeast Asian cities have fragmented app ecosystems.

Worldwide comparison of mobility app market concentration and mode coverage



2. Influence of individual attributes on route search behavior

- Bus users switch more frequently apps than taxi users.
- Older users have higher dwell time than younger group, and short-term residents have fewer app switches.

Estimated coefficients of parameters in the SUR model

Variables (dummies)	App-switch counts	In-app dwell time
High-Age (>25 years)	-0.145	0.252 *
Bus Users	0.562 *	-0.040
Taxi Users	-0.556 **	-0.179
Shorter living period (<10 years)	-0.296 *	0.022

*: p<0.05, **: p<0.01

5 Conclusions

- Developing Asian cities have a fragmented mobility app market, imposing a searching barrier for vulnerable populations.
- Individuals' searching strategy reflects their frequently used transport modes.
- Both policy and developers' intervention are important to integrate information and reduce searching burdens.

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Final Study
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program in 2026
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